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ICE 9A.2



Aman Gupta • Apr 10, 2025 (Edited Apr 27, 2025)

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1. (2 marks) Consider the non-linear 3-input, 1 output logical expression $[(A \text{ and } B) \text{ or } (\text{not } A \text{ and } C)]$, or equivalently $[\text{if } A \text{ then } B \text{ else } C]$. If the output is 0, then what is (A, B, C)?

- (0 0 0)
- (0 1 0)
- (0 1 1)
- (1 0 0)
- (1 0 1)
- (1 0 1), or (0 1 1), or (0 0 0)
- (0 0 0), or (0 1 0), or (1 0 0), or (1 0 1) (correct)
- None of the above

2. (2 marks) Consider the non-linear 3-input, 1 output logical expression $[(A \text{ and } B) \text{ or } (A \text{ and } C) \text{ or } (B \text{ and } C)]$, or equivalently $[\text{majority of } A, B, C]$. If the output is 1, then what is (A, B, C)?

- (0 1 0)
- (0 1 1)
- (1 0 1)
- (1 1 0)
- (0 1 1) or (1 0 1) or (1 1 0) or (1 1 1) (correct)
- (0 0 0) or (0 1 0) or (1 0 0)
- None of the above

3. (2 marks) What is the time complexity of an attack on 1st pre-image resistance of SHA512. It is known that SHA512 operates on blocks of $m=1024$ message bits to generate $n=512$ bits hash.

- Requires 2^m attempts (m is length of message)
- Requires 2^n attempts (n is length of hash) (correct)
- None of the above
- Cannot say

4. (2 marks) What is the time complexity of an attack on 2nd pre-image resistance of SHA512 as compared to an attack on 1st pre-image resistance of SHA512?

- Complexity of an attack on 2nd pre-image is much MORE complex
- Complexity of an attack on 2nd pre-image is LESS complex
- Complexity of an attack on 2nd pre-image is the same as an attack on the 1st pre-image (correct) 

- None of the above



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Your work

Graded

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